WoSM Academy

MAT231-F27 · Calculus II

Systems of linear differential equations

Eigenvalues, phase portraits, and the matrix exponential

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Last time

- $ay'' + by' + cy = 0$: the characteristic equation $ar^2 + br + c = 0$; real, double and complex roots; Euler's formula.
- Free and damped oscillations; forced oscillations and resonance.

Today: several unknown functions at once, $\mathbf{x}' = A\mathbf{x}$. The characteristic equation becomes the eigenvalue equation $\det(A - \lambda I) = 0$ — and Linear Algebra and Calculus meet.

This lecture

01 Linear systems

02 Solutions from eigenvectors

03 Phase portraits and stability

04 The matrix exponential

05 Looking back

06 Summary and practice

Reading: Apostol, *Calculus, Volume 2*, Chapter 7.

01

Linear systems

Systems of differential equations

A **linear system with constant coefficients**: $\mathbf{x}'(t) = A\,\mathbf{x}(t)$, with $\mathbf{x}(t) \in \mathbb{R}^n$ and A a constant $n \times n$ matrix.

Example (Two rooms)

Two rooms lose heat to the outside, and exchange heat with each other, at rate 1. With x_1, x_2 their temperatures above the outside temperature:

$$x_1' = -x_1 + (x_2 - x_1), \quad x_2' = -x_2 + (x_1 - x_2), \qquad A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}.$$

A second-order equation is a system

For $y'' + by' + cy = 0$, put $x_1 = y$ and $x_2 = y'$. Then $x_1' = x_2$ and $x_2' = -cx_1 - bx_2$:

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -c & -b \end{pmatrix} \mathbf{x}, \qquad \det(A - \lambda I) = \lambda^2 + b\lambda + c,$$

the characteristic polynomial of Lecture 24.

- In the same way, any equation of order n becomes a first-order system in $\mathbb{R}^n$.
- Numerical solvers (Euler's method and its refinements) work with first-order systems, so this is how higher-order equations are solved on a computer.

Existence, uniqueness, superposition

Theorem

For every $\mathbf{x}_0 \in \mathbb{R}^n$, $\mathbf{x}' = A\mathbf{x}$ has exactly one solution with $\mathbf{x}(0) = \mathbf{x}_0$, and it is defined for all t.

- Sums and multiples of solutions are solutions, and $\mathbf{x} \mapsto \mathbf{x}(0)$ is a linear bijection onto $\mathbb{R}^n$: the solutions form an n-dimensional vector space.
- So we need n independent solutions; every solution is a combination of them.
- Geometrically, solutions are the flow lines of the vector field $\mathbf{x} \mapsto A\mathbf{x}$ (Lecture 22). Their pictures in the (x_1, x_2)-plane form the **phase portrait**.

02

Solutions from eigenvectors

Straight-line solutions

Try $\mathbf{x}(t) = e^{\lambda t}\,\mathbf{v}$ with a constant vector $\mathbf{v} \neq \mathbf{0}$:

$$\mathbf{x}' = \lambda e^{\lambda t}\mathbf{v}, \qquad A\mathbf{x} = e^{\lambda t} A\mathbf{v}.$$

So $e^{\lambda t}\mathbf{v}$ is a solution exactly when $A\mathbf{v} = \lambda\mathbf{v}$: $\mathbf{v}$ is an **eigenvector** and λ an **eigenvalue** of A (Linear Algebra, Unit 17).

- The solution moves along the line through $\mathbf{v}$: outwards if $\lambda > 0$, inwards if $\lambda < 0$.
- The eigenvalues are the roots of $\det(A - \lambda I) = 0$, the analogue of the characteristic equation.

The general solution

Theorem

If A has n linearly independent eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ with eigenvalues $\lambda_1, \ldots, \lambda_n$, every solution of $\mathbf{x}' = A\mathbf{x}$ is

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t}\mathbf{v}_1 + \cdots + c_n e^{\lambda_n t}\mathbf{v}_n,$$

and the c_i are found from $\mathbf{x}(0) = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$.

Why: with $V = (\mathbf{v}_1 \cdots \mathbf{v}_n)$, $A = V\Lambda V^{-1}$ (Linear Algebra, Unit 18). Put $\mathbf{x} = V\mathbf{y}$: then $\mathbf{y}' = \Lambda\mathbf{y}$, that is $y_i' = \lambda_i y_i$ for each i — n separate equations of Lecture 23, with $y_i = c_i e^{\lambda_i t}$.

Eigenvectors **decouple** the system. Each eigendirection evolves on its own.

Example: two rooms

$A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$: $\det(A - \lambda I) = (\lambda + 2)^2 - 1$, so $\lambda = -1$ with $\mathbf{v} = (1, 1)$, and $\lambda = -3$ with $\mathbf{v} = (1, -1)$.

Start with room 1 warm, room 2 not: $\mathbf{x}(0) = (1, 0) = \frac{1}{2}(1, 1) + \frac{1}{2}(1, -1)$. So

$$\mathbf{x}(t) = \frac{1}{2} e^{-t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} e^{-3t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

- The **difference** $x_1 - x_2 = e^{-3t}$ disappears fast: the rooms equalise.
- The **average** $\frac{1}{2}(x_1 + x_2) = \frac{1}{2} e^{-t}$ decays slowly: both cool together.

Complex eigenvalues

If $\lambda = \alpha + i\beta$ ($\beta \neq 0$) is an eigenvalue of a real matrix A, with eigenvector $\mathbf{v}$, the real and imaginary parts of $e^{\lambda t}\mathbf{v}$ are two real solutions (as in Lecture 24).

Example. $A = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix}$: $\det(A - \lambda I) = (\lambda + 1)^2 + 4$, so $\lambda = -1 \pm 2i$. For $\lambda = -1 + 2i$, $\mathbf{v} = (1, i)$:

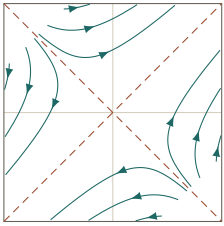
$$e^{\lambda t}\mathbf{v} = e^{-t}(\cos 2t + i \sin 2t) \begin{pmatrix} 1 \\ i \end{pmatrix} = e^{-t} \begin{pmatrix} \cos 2t \\ -\sin 2t \end{pmatrix} + i\, e^{-t} \begin{pmatrix} \sin 2t \\ \cos 2t \end{pmatrix}.$$

The solution with $\mathbf{x}(0) = (1, 0)$ is $e^{-t}(\cos 2t, -\sin 2t)$: it turns clockwise and shrinks — a **spiral**.

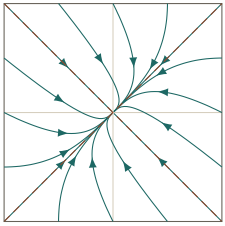
03

Phase portraits and stability

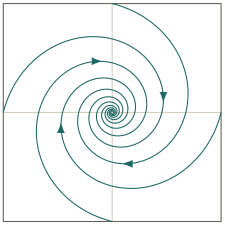
Phase portraits



Saddle:
$\lambda = 3, -1.$
$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$



Stable node:
$\lambda = -1, -3.$
$A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$



Stable spiral:
$\lambda = -\frac{1}{4} \pm i.$
$A = \begin{pmatrix} -1/4 & 1 \\ -1 & -1/4 \end{pmatrix}$

Dashed: the eigenvector lines, along which solutions move straight.

Classification by trace and determinant

For a 2×2 matrix, $\lambda_1 + \lambda_2 = \operatorname{tr} A = T$ and $\lambda_1 \lambda_2 = \det A = D$, so $\lambda = \frac{1}{2}\left(T \pm \sqrt{T^2 - 4D}\right)$.

Condition	Eigenvalues	Origin
$D < 0$	real, opposite signs	saddle
$D > 0,\ T^2 \geq 4D$	real, same sign	node
$D > 0,\ T^2 < 4D,\ T \neq 0$	complex, $\operatorname{Re}\lambda = T/2$	spiral
$D > 0,\ T = 0$	purely imaginary	centre (closed orbits)

Nodes and spirals are **stable** (attracting) if $T < 0$ and unstable if $T > 0$. Example: the damped spring $\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -k/m & -c/m \end{pmatrix} \mathbf{x}$ has $T = -c/m < 0$ and $D = k/m > 0$: stable, a spiral exactly when underdamped.

Stability

Theorem

Every solution of $\mathbf{x}' = A\mathbf{x}$ tends to $\mathbf{0}$ as $t \to \infty$ if and only if every eigenvalue of A has negative real part.

(For diagonalisable A: from $\sum_i c_i e^{\lambda_i t}\mathbf{v}_i$ and $|e^{\lambda t}| = e^{(\operatorname{Re}\lambda)\,t}$.)

Linearisation (outlook)

Near an equilibrium $\mathbf{x}^*$ of a nonlinear system $\mathbf{x}' = F(\mathbf{x})$, the Jacobian $DF(\mathbf{x}^*)$ (Lecture 12) usually decides stability.

Example: the gradient flow $\mathbf{x}' = -\nabla f(\mathbf{x})$ has Jacobian $-H_f$. At a strict local minimum H_f is positive definite (Lecture 14), so the minimum attracts. Gradient descent is Euler's method for this flow.

04

The matrix exponential

The matrix exponential

Definition

For a square matrix A: $\quad e^{tA} = I + tA + \dfrac{t^2A^2}{2!} + \dfrac{t^3A^3}{3!} + \cdots$

- The series converges for every A and t: each entry of $t^kA^k/k!$ is at most $(|t|\,nM)^k/k!$ in absolute value, where M is the largest $|a_{ij}|$ — compare with the series of $e^{|t|nM}$ (Lectures 2 and 5).
- Differentiating term by term (Lecture 5): $\frac{d}{dt}e^{tA} = Ae^{tA}$. So

$$\mathbf{x}(t) = e^{tA}\,\mathbf{x}_0$$

solves $\mathbf{x}' = A\mathbf{x}$, $\mathbf{x}(0) = \mathbf{x}_0$ — exactly like $x(t) = e^{at}x_0$ in one variable.

Computing e^{tA}

- **Diagonalisable**: $A = V\Lambda V^{-1}$ gives $A^k = V\Lambda^k V^{-1}$, so

$$e^{tA} = V\ \mathrm{diag}(e^{\lambda_1 t}, \ldots, e^{\lambda_n t})\, V^{-1}.$$

  Then $e^{tA}\mathbf{x}_0$ is the eigenvector solution from before.

- **A rotation**: $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ has $A^2 = -I$. Separating even and odd powers, as for Euler's formula:

$$e^{tA} = I\cos t + A\sin t = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}.$$

When A is not diagonalisable

$A = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} = \lambda I + N$, with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $N^2 = 0$, has only one eigenvector direction.

Since λI and N commute, $e^{t(\lambda I + N)} = e^{\lambda t}\, e^{tN}$, and the series for e^{tN} stops after two terms:

$$e^{tA} = e^{\lambda t}\,(I + tN) = e^{\lambda t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

The factor $te^{\lambda t}$ is the double root of Lecture 24 again. In general every matrix has a Jordan form (Linear Algebra, Unit 24), and e^{tA} contains terms $t^k e^{\lambda t}$.

05

Looking back

The module in one slide

- **Series** (Units 1–8): convergence tests, power and Taylor series, generating functions, Stirling's formula.
- **Differential calculus in several variables** (Units 9–17): gradients, the chain rule and Jacobians, the Hessian and Taylor's formula, optimisation with and without constraints, gradient descent, implicit and inverse functions.
- **Integration** (Units 18–22): double and triple integrals, change of variables, the Gaussian integral, line integrals.
- **Differential equations** (Units 23–25): first-order, second-order linear, and linear systems.

Some threads: power series gave $e^{i\theta}$ and e^{tA}; eigenvalues decided the Hessian test, the speed of gradient descent, Lagrange problems on the sphere, and stability today; the Jacobian determinant governed the chain rule, inverse functions and the change of variables.

06

Summary and practice

Summary

- $\mathbf{x}' = A\mathbf{x}$: unique solutions for all t; an n-dimensional space of solutions.
- Eigenpairs give solutions $e^{\lambda t}\mathbf{v}$; with n independent eigenvectors, $\mathbf{x} = \sum c_i e^{\lambda_i t}\mathbf{v}_i$. Complex eigenvalues: take real and imaginary parts.
- 2×2 phase portraits: saddle, node, spiral, centre; classified by $\operatorname{tr} A$ and $\det A$. Stable iff all eigenvalues have negative real part.
- The matrix exponential: $\mathbf{x}(t) = e^{tA}\mathbf{x}_0$, with $e^{tA} = Ve^{t\Lambda}V^{-1}$ when A is diagonalisable.

Practice

1. Solve $\mathbf{x}' = A\mathbf{x}$ with $A = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$ and $\mathbf{x}(0) = (3, 2)$.
2. Classify the origin for $\begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$, $\begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$, $\begin{pmatrix} 1 & -5 \\ 1 & -1 \end{pmatrix}$.
3. Solve $x' = y$, $y' = -4x$ with $x(0) = 1$, $y(0) = 0$. What curve does $(x(t), y(t))$ trace?
4. Write $y'' + 3y' + 2y = 0$ as a system, and compare the eigenvalues of its matrix with the roots of the characteristic equation.
5. Compute e^{tA} for $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, and solve $\mathbf{x}' = A\mathbf{x}$.
6. For which k do all solutions of $\mathbf{x}' = \begin{pmatrix} -1 & k \\ 1 & -1 \end{pmatrix}\mathbf{x}$ tend to $\mathbf{0}$?
7. Show directly that if $A\mathbf{v} = \lambda\mathbf{v}$, then $e^{\lambda t}\mathbf{v}$ solves $\mathbf{x}' = A\mathbf{x}$.

Answers

1. $\lambda = 2$, $\mathbf{v} = (1,1)$; $\lambda = 3$, $\mathbf{v} = (2,1)$; $(3,2) = (1,1) + (2,1)$: $\mathbf{x} = e^{2t}(1,1) + e^{3t}(2,1)$.
2. $D = -4$: saddle. $T = -6$, $D = 8$, $T^2 > 4D$: stable node. $T = 0$, $D = 4$: centre.
3. $x = \cos 2t$, $y = -2\sin 2t$: the ellipse $x^2 + \frac{y^2}{4} = 1$, traversed clockwise.
4. $\begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$: $\lambda^2 + 3\lambda + 2 = 0$, so $\lambda = -1, -2$ — the same as the roots of $r^2 + 3r + 2 = 0$.
5. $A^2 = 0$: $e^{tA} = I + tA = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$; $x_1 = x_1(0) + t\,x_2(0)$, $x_2 = x_2(0)$.
6. $T = -2 < 0$; need $D = 1 - k > 0$: $k < 1$.
7. $(e^{\lambda t}\mathbf{v})' = \lambda e^{\lambda t}\mathbf{v} = e^{\lambda t}A\mathbf{v} = A(e^{\lambda t}\mathbf{v})$.

The end of the module

This was the last unit of Calculus II.

- **Final exam**: written, 3 hours, at the end of term (45% of the module). It tests computation (series, derivatives and integrals in several variables), the choice of method, and short proofs about convergence.
- **What uses this module**: multiple integrals and the Gaussian integral in Probability Theory and Statistics (joint densities); series and generating functions in the analysis of algorithms; gradients, Hessians and gradient descent in optimisation and machine learning.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.